Mandatory deliverable

LAB | Create a CloudWatch Alarm Lab

Introduction

In this hands-on lab, you will create an Amazon CloudWatch alarm to monitor the CPU utilization of an EC2 instance. If the CPU usage exceeds a predefined threshold, the alarm will trigger a notification via Amazon Simple Notification Service (SNS).

Objectives

By the end of this lab, you will be able to:

1. Launch an EC2 instance and retrieve its instance ID.
2. Create a CloudWatch alarm to monitor CPU utilization.
3. Configure SNS notifications to receive alerts when the alarm is triggered.
4. Simulate high CPU usage using the `stress` utility.
5. Verify the alarm activation and notification delivery.
6. Clean up AWS resources to prevent unnecessary charges.

Lab Instructions

1. Create an EC2 Instance

- Log in to your AWS account.
- Navigate to the **EC2 Dashboard** and launch a new instance with the following settings:
  - **AMI**: Amazon Linux 2
  - **Instance Type**: `t2.micro`
  - **Storage**: Default 8GB root volume
  - **Security Group**: Allow SSH (port 22) from your IP
- **Launch the instance** and **note down** the **Instance ID**.

2. Create a CloudWatch Alarm

Ask AI Tutor

- Go to the **CloudWatch Console** → **Alarms** → **Create Alarm**.
- Click on **Select metric** → **EC2 Metrics** → **Per-Instance Metrics**.
- Choose **CPUUtilization** and select the previously created **EC2 instance**.
- Set the following parameters:
  - **Statistic**: Average
  - **Period**: 5 minutes
  - **Threshold Type**: Static
  - **Alarm Condition**: Greater than `70%`
- Click **Next**.

3. Configure SNS Notifications

- **Create a new SNS topic**:
  - **Topic name**: `cpu-alert`
  - **Create a subscription** → **Email** → Enter your email address.
  - **Confirm the subscription** via the verification email sent to your inbox.
- Assign this SNS topic to the **CloudWatch Alarm Notification Settings**.
- Name the alarm (e.g., `cpu-test`) and **Create Alarm**.

4. Simulate High CPU Usage

- Connect to the EC2 instance using SSH:

```
ssh -i your-key.pem ec2-user@<public-ip>
```

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- Install the stress utility:

```
sudo yum -y install stress
```

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- Run a CPU load simulation for 5 minutes:

```
stress --cpu 1
```

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5. Verify Alarm Activation and Notification

- Return to **CloudWatch** and check the **Alarm Status**.
- It should change from **OK** → **In Alarm** within a few minutes.
- Check your email inbox for the notification triggered by the CloudWatch alarm.

6. Cleanup Resources

To avoid unnecessary charges, **delete all resources** after completing the lab:

- **Terminate the EC2 instance** (EC2 Console → Instances → Terminate).
- **Delete the CloudWatch Alarm** (CloudWatch Console → Alarms → Delete).
- **Remove the SNS topic** (SNS Console → Topics → Delete `cpu-alert`).

Completion Criteria

- EC2 instance is launched and monitored
- CloudWatch alarm is created and configured
- SNS topic is set up and email notifications received
- CPU stress test successfully triggers the alarm
- All resources are cleaned up to prevent charges

Submission Guidelines

- **Take screenshots** of the following:
  - CloudWatch alarm configuration
  - SNS email notification
  - SSH session running `stress` command
  - Alarm state change to **"In Alarm"**
- **Compile screenshots into a document** (Google Docs, PDF, etc.).
- **Upload to Google Drive** and share the link with your instructor.

🚀 **Great job! You've successfully created and tested a CloudWatch alarm!** 🎉

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Lab

URL

Enter the URL here

Submit lab